॥ श्रीगणेशाय नमः ॥

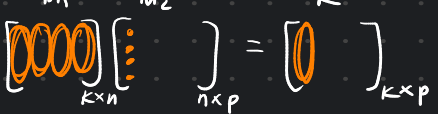
LU Decomposition

Perspectives on matrix multiplication

1st perspective:
a_{ij} of resultant matrix (R) is dot product betn i^{th} row of 1st matrix with j^{th} colm of 2nd matrix.

2nd perspective:
Each colm of R is a linear combn of colms of M_1 weighted by each colm of M_2.

(i^{th} colm of R) M_1 M_2 R (i^{th} colm of M_2)



M_1 : $K \times n$, M_2 : $n \times p$, R : $K \times p$

Each colm of R is in the colm space of M_1.

3rd perspective:
Each row of R is a linear combn of rows of M_2 weighted by rows of M_1

(i^{th} row of R) (i^{th} row of M_1)



Example:

$$\begin{bmatrix}1 & 2\end{bmatrix}_{1\times 2}\begin{bmatrix}3 & 4\\ 5 & 6\end{bmatrix}_{2\times 2} = \begin{bmatrix}\quad & \quad\end{bmatrix}_{1\times 2}$$

each colm is linear combn of colms of M_2

$c_1 = 1(3)+2(5) = 13$ $\quad c_2 = 1(4)+2(6) = 16$ $\quad \begin{bmatrix}13 & 16\end{bmatrix}$

3rd perspective:

$$\begin{matrix}1(3,4)\\ +2(5,6)\end{matrix} = \begin{matrix}3,4\\ +10,12\end{matrix} = \begin{bmatrix}13 & 16\end{bmatrix}$$

Elimination matrix

Simple system:

$$\begin{aligned} x+y &= 2\\ x-y &= 2\\ \hline 2x &= 4\\ x &= 2\\ y &= 0\end{aligned}$$

← we eliminate y. can we perform these operars using matrices?

$$\begin{aligned} 2x-y-z &= 8\\ -3x-y+2z &= -11\\ -2x+y+2z &= -3\end{aligned}$$

A|b:

$$\left[\begin{array}{ccc|c} 2 & -1 & -1 & 8\\ -3 & -1 & 2 & -11\\ -2 & 1 & 2 & -3\end{array}\right]$$

we want to get closer to identity matrix $I_{3\times 3}$ (as in RREF)

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8\\ -3 & -1 & 2 & -11\\ -2 & 1 & 2 & -3\end{array}\right]$$

consider a simple example:

$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} M_1$

$R_2 \leftarrow R_2 - R_1$

$\begin{pmatrix} 1 & 1 \\ 0 & -2 \end{pmatrix} E_1 M_1$

Representing as matrix:

same operation on I: $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$

↳ This is the matrix that represents that operation.

(E_1)

$$E_1 M_1 = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & -2 \end{pmatrix}$$

(we multiply $E \cdot M$ and not $M \cdot E$ as in $A \times B$, row space of A B is preserved)

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \xrightarrow{E_1} \begin{pmatrix} 1 & 1 \\ 0 & -2 \end{pmatrix} \xrightarrow{E_2} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

$R_2 \leftarrow R_2 \times (\frac{-1}{2})$

Doing same to I:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1/2 \end{pmatrix}$$

E_2

$$\underset{M_1}{\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}} \xrightarrow{E_1} \underset{M_2}{\begin{pmatrix} 1 & 1 \\ 0 & -2 \end{pmatrix}} \xrightarrow{E_2} \underset{M_3}{\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}} \xrightarrow{E_3} \underset{M_4 = I}{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}$$

$R_1 \leftarrow R_1 - R_2$

on I: $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$ E_3

$$E_3 E_2 E_1 M_1 = I \qquad E_1 = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \quad E_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1/2 \end{pmatrix} \quad E_3 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

$E = E_1 E_2 E_3$ ⟶ we are only bothered by row space so order of multiplication doesn't matter, row space won't change.

$E M_1 = I$

$M_1^{-1} = E$

(A)

So if matrix is invertible ⇒ unique soln exists. (for $Ax = b$)

(As getting to I means we can get a unique soln)

Invertible matrices:

- If there exists an elimination matrix for a given M s.t. $EM = I$, then M is invertible.

Q: Find elimination matrix:

$$\begin{bmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{bmatrix}$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ -3 & -1 & 2 & 0 & 1 & 0 \\ -2 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

$R_1 \leftarrow R_1 \times (\frac{1}{2})$, $R_2 \leftarrow (-1) R_2$, $R_3 \leftarrow (-1) R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & -1/2 & 1/2 & 0 & 0 \\ 3 & 1 & -2 & 0 & -1 & 0 \\ 2 & -1 & -2 & 0 & 0 & -1 \end{array}\right]$$

$R_2 \leftarrow R_2 - 3R_1$, $R_3 \leftarrow R_3 - 2R_1$

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & -1/2 & 1/2 & 0 & 0 \\ 0 & -1/2 & -1/2 & -3/2 & -1 & 0 \\ 0 & -2 & -1 & -1 & 0 & -1 \end{array}\right]$$

$R_2 \leftarrow -2 \times R_2$ $R_3 \leftarrow (-1) R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & -1/2 & 1/2 & 0 & 0 \\ 0 & 1 & 1 & 3 & 2 & 0 \\ 0 & 2 & 1 & 1 & 0 & 1 \end{array}\right]$$

$R_1 \leftarrow R_1 - 1/2\, R_2$ $R_3 \leftarrow R_3 - 2R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -1 & -1 & -1 & 0 \\ 0 & 1 & 1 & 3 & 2 & 0 \\ 0 & 0 & -1 & -5 & -4 & 1 \end{array}\right]$$

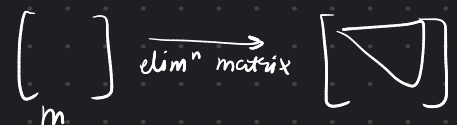
$R_1 \leftarrow R_1 - R_3$ $R_2 \leftarrow R_2 + R_3$ $R_3 \leftarrow R_3 (-1)$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 4 & 3 & -1 \\ 0 & 1 & 0 & -2 & -2 & 1 \\ 0 & 0 & 1 & 5 & 4 & -1 \end{array}\right]$$

→ Inverse matrix

Elimination matrix

We want to first reach some upper bar matrix (row-echelon form) 



We can get soln from upper bar matrix as well.

$$\begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$z = \frac{b_3}{f}$ $y = \frac{b_2 - e\frac{b_3}{f}}{d}$ $x = \dots$

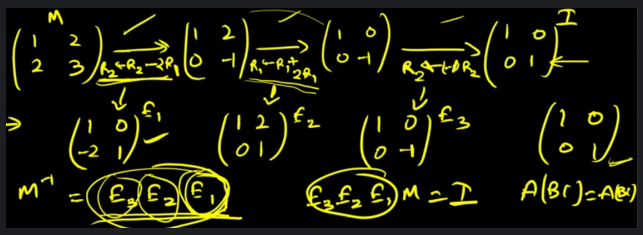
More properties of row operations

- Row operations are invertible.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \xrightarrow{E_1} \begin{pmatrix} a & b \\ 0 & d-\frac{c}{a}b \end{pmatrix}$$

$$E_1 = \begin{pmatrix} 1 & 0 \\ -\frac{c}{a} & 1 \end{pmatrix} \qquad E_1^{-1} : R_2 \leftarrow R_1 + \frac{c}{a} R_1$$

e.g.



$$E_3 E_2 E_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} -3 & 2 \\ 2 & -1 \end{pmatrix}$$

Row operations: $R^1 : R_2 \leftarrow R_2 - 2R_1$, $R^2 : R_1 \leftarrow R_1 + 2R_2$

$R^3 : R_2 \leftarrow (-1)R_2$

Inverse of row operations: $(R^1)^{-1} : R_2 \leftarrow R_2 + 2R_1$

$(R^2)^{-1} : R_1 \leftarrow R_1 - 2R_2$

$(R^3)^{-1} : R_2 \leftarrow (-1)R_2$

$$E_1^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \quad E_2^{-1} = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} \quad E_3^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$E_1^{-1} \cdot E_2^{-1} E_3^{-1} = M$$

Finding inverse using augmented matrix:

$$[M \mid I] = \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 3 & 0 & 1 \end{array}\right)$$

$R_2 \leftarrow R_2 - 2R_1$

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -1 & -2 & 1 \end{array}\right)$$

$R_1 \leftarrow R_1 + 2R_2$

$$\left(\begin{array}{cc|cc} 1 & 0 & -3 & 2 \\ 0 & -1 & -2 & 1 \end{array}\right)$$

$R_2 \leftarrow -R_2$

$$\left(\begin{array}{cc|cc} 1 & 0 & -3 & 2 \\ 0 & 1 & 2 & -1 \end{array}\right)$$

Properties of matrix inverses

- Inverse of a matrix exists iff the columns (or rows) of the matrix are LI.

as we need n pivots.

$(A_{m \times n})$

- Inverse of a matrix exists iff there are n pivots in its echelon form.
- Only square matrices can have inverses.
- If we have some non-zero solution to Ax=0, then A is not invertible.

$Ax = 0$

- unique (zero)
- only many (non-zero) → $< n$ pivots

- Inverse as a function follows one-to-one mapping.

$Ax = b \quad \Rightarrow A^{-1}Ax = A^{-1}b$

$x = A^{-1}b$

← If we have multiple options for $A^{-1} \Rightarrow A^{-1}$ doesn't exist.

- Inverse of a diagonal matrix is also a diagonal matrix.

$(M|I) = \left(\begin{array}{cc|cc} a & 0 & 1 & 0 \\ 0 & b & 0 & 1 \end{array}\right)$

$R_1 \leftarrow \frac{1}{a} R_1 \quad R_2 \leftarrow \frac{1}{b} R_2$

$\left(\begin{array}{cc|cc} 1 & 0 & 1/a & 0 \\ 0 & 1 & 0 & 1/b \end{array}\right)$

$= M^{-1}$

- Inverse of a △ar matrix is also a △ar matrix.

$\left(\begin{array}{cc|cc} a & b & 1 & 0 \\ 0 & c & 0 & 1 \end{array}\right)$

$\left(\begin{array}{cc|cc} 1 & b/a & 1/a & 0 \\ 0 & 1 & 0 & 1/c \end{array}\right)$

$\left(\begin{array}{cc|cc} 1 & 0 & \frac{1}{a} & -\frac{b}{ac} \\ 0 & 1 & 0 & \frac{1}{c} \end{array}\right)$

LU decomposition

- It is a matrix factorization method to decompose a matrix A into
  a. a lower △ar matrix L.
  and b. an upper △ar matrix U.

$A = LU$

To solve $Ax = b$, we perform row operations (E) to reach echelon form (upper △ar)

$Em = U$

lower △ar

$m = E^{-1}U$

LU

upper △ar

$A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} \xrightarrow[R_2 \leftarrow R_2 - 2R_1]{} \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$

$E_1 = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$

$E_1 m = U$

$m = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}^{-1} U$

$\hookrightarrow L$

Restriction on L:
Diagonals must be 1.

$\left(\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ -2 & 1 & 0 & 1 \end{array}\right) \rightarrow \left(\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 \end{array}\right)$

$m = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$

$m = LU$ ↳ we get diagonals 1 automatically.

Q. $A = \begin{pmatrix} 4 & 3 \\ 6 & 3 \end{pmatrix}$

$\rightarrow \begin{pmatrix} 4 & 3 \\ 6 & 3 \end{pmatrix} \rightarrow \begin{pmatrix} 4 & 3 \\ 2 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$

$E_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1/3 \end{pmatrix}$ $E_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $E_3 = ($

$\begin{pmatrix} 4 & 3 \\ 6 & 3 \end{pmatrix} \xrightarrow[R_2 \leftarrow R_2 - \frac{3}{2}R_1]{} \begin{pmatrix} 4 & 3 \\ 0 & -1.5 \end{pmatrix}$

$E = \begin{pmatrix} 1 & 0 \\ -\frac{3}{2} & 1 \end{pmatrix}$ $L = E^{-1}$ $\left(\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ -1.5 & 1 & 0 & 1 \end{array}\right) \rightarrow \left(\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 1 & 1.5 & 1 \end{array}\right)$

$\begin{pmatrix} 4 & 3 \\ 6 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1.5 & 1 \end{pmatrix}\begin{pmatrix} 4 & 3 \\ 0 & -1.5 \end{pmatrix}$

How to find L faster?
Follow order for row operations to find U.

$\begin{pmatrix} 4 & 3 \\ 6 & 3 \end{pmatrix} \xrightarrow[R_2 \leftarrow R_2 - \frac{3}{2}R_1]{} \begin{pmatrix} 4 & 3 \\ 0 & -1.5 \end{pmatrix}$

Instead of finding E & then E^{-1}, we just take inverse of row operations & take matrix.

$R_2 \leftarrow R_2 + \frac{3}{2}R_1$

$\begin{pmatrix} 1 & 0 \\ 1.5 & 1 \end{pmatrix}$

Q. $\begin{pmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{pmatrix}$ —(all zero below 1st pivot)→ ① $R_2 \leftarrow R_2 - 2R_1$, ② $R_3 \leftarrow R_3 + R_1$ → $\begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 8 & 3 \end{pmatrix}$ —(all zero below 2nd pivot)→ ③ $R_3 \leftarrow R_3 + R_2$ → $\begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{pmatrix}$

we could find E_1, E_2, E_3 and then take inverses.

OR...

Instead we could take inverses of row operations.

① $R_2 \leftarrow R_2 + 2R_1$

② $R_3 \leftarrow R_3 - R_1$

③ $R_3 \leftarrow R_3 - R_2$

$E_3 E_2 E_1 M = U$

$M = E_1^{-1} E_2^{-1} E_3^{-1} U$

$L = E_1^{-1} E_2^{-1} E_3^{-1}$

L will be lower Δr with diagonals 1.

$\begin{pmatrix} 1 & 0 & 0 \\ \square & 1 & 0 \\ \square & \square & 1 \end{pmatrix}$

Apply inverse of 3rd op: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$ $R_3 = R_3 - R_2$

2nd: $R_3 = R_3 - R_1$

$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix}$

$R_2 = R_2 + 2R_1$

$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix}$ ← L

$Ax = b$

$x = A^{-1}b$ ← we could use LU decomposition here

→ $A = LU$

$LUx = b$

$Ux = (L^{-1}b)_{n\times 1}$

$Ux = s$ ← some vector

we get another system.

$L^{-1}b = S$

$b = LS$

b is in the col^m space of L.

↳ same as col^m space of A.

HW2:

$$A = \begin{pmatrix} 3 & 4 & 6 & 0 \\ 1 & 2 & 2 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$